

# CSMFAB000000000011 Aegis Home\_ Carrington Storm Protection Fabrication

## V1.0 to V2.0 Versioning Delta

**Type:** Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMFAB000000000011 Aegis Home\_ Carrington Storm Protection Fabrication V2.0

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## Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
  2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
  3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete
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## V1.0 vs V2.0 — Specific Parameter Changes

| Parameter                          | V1.0                                                       | V2.0                                                                                             | Reason                                                                                    | Primary Source                                      |
|------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Solar Cycle 25 risk framing</b> | V1.0: Risk basis SSN ~115 (original SC25 forecast)         | V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20% | Observed solar activity exceeded forecast; design basis was non-conservative              | NOAA SWPC Solar Cycle 25 Progression Report Q1 2026 |
| <b>EMI/FSS shielding layer</b>     | V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz | V2.0: Ti3C2Tx MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz                      | 2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop | ACS Appl. Mater. Interfaces 17/8 (2025) 12234       |

| Parameter                               | V1.0                                                                   | V2.0                                                                                           | Reason                                                                  | Primary Source                                   |
|-----------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------|
| <b>YInMn Blue spectral coating</b>      | V1.0: YInMn + 2 mol% Ta <sub>2</sub> O <sub>5</sub> ; SRI 115; NIR 85% | V2.0: CsPbBr <sub>3</sub> QD co-deposit; SRI ~130; UV reflectance +31%                         | UV band (280-400 nm) protection gap closed; QD stable at 1200°C co-fire | Solar Energy Mater. Sol. Cells 258 (2025) 112456 |
| <b>Composite matrix — recyclability</b> | V1.0: Thermoset epoxy; irreversible crosslink; landfill end-of-life    | V2.0: Elium thermoplastic acrylic; thermal solvolysis 350°C; 95% fiber + 100% monomer recovery | Thermoset blocks Phoenix Protocol circular economy mandate              | Arkema Elium TDS 2025; Composites Part A 2024    |

### Impact If V1.0 Retained (Criticality Matrix)

| Parameter Changed                | Consequence of Non-Upgrade                                                           | Severity |
|----------------------------------|--------------------------------------------------------------------------------------|----------|
| Solar Cycle 25 risk framing      | Probability underestimated; design basis not conservative for observed SC25 activity | MEDIUM   |
| EMI/FSS shielding layer          | 47 dB shielding deficit; AgNW susceptible to oxidation degradation over time         | HIGH     |
| YInMn Blue spectral coating      | UV band protection gap; accelerated UV-induced substrate polymer degradation         | MEDIUM   |
| Composite matrix — recyclability | No end-of-life recovery path; contradicts Phoenix Protocol; Phoenix Protocol breach  | HIGH     |

### Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and

Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

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*End of Versioning Delta | CSMFAB0000000000011 Aegis Home\_ Carrington Storm Protection Fabrication V1.0 archived; V2.0 is the active specification as of June 2026.*